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ASSOCIATION OF BODY MASS INDEX CHANGES WITH CHRONIC KIDNEY DISEASE IN A JAPANESE GENERAL POPULATIONAkiko Toda¹, Shigeko Hara¹, Hiroshi Tsuji¹, Yasuji Arase¹¹Toranomon Hospital, Health Management Center, Tokyo, Japan

Background and Aims: Obesity is a risk factor for chronic kidney disease (CKD), but the effect of reducing body mass index (BMI) on the prevention of CKD is controversial. One of reasons for this disagreement is that part of patients with a BMI decrease may have an unfavourable health status. In such cases, the BMI decrease could be a risk factor for the development of CKD. Therefore, by analysing the data of annual health check-ups, we examined an association between BMI change and CKD development to determine whether BMI reduction helps prevent CKD development.

Method: We analysed the data of 6,959 subjects who underwent annual health check-ups in both 2013 and 2018. By a multivariate logistic regression analysis, we investigated a relationship between BMI change and CKD development within the 5 years between 2013 and 2018. The percent change in the BMI (Δ BMI) was calculated using the following equation: $\{(\text{BMI in 2018} - \text{BMI in 2013})/\text{BMI in 2013}\} \times 100$. For analyses, we classified the subjects into five groups based on their Δ BMI value: (i) severe BMI decrease (Δ BMI $< -2.5\%$); (ii) moderate BMI decrease (Δ BMI $\geq -2.5\%$ but $< 0\%$); (iii) maintained BMI (Δ BMI $\geq 0\%$ but $< 2.5\%$); (iv) moderate BMI increase (Δ BMI $\geq 2.5\%$ but $< 5\%$); (v) severe BMI increase (Δ BMI $\geq 5\%$). For further analysis, we divided the subjects into non-obesity category (basal BMI $< 25 \text{ Kg/m}^2$) and obesity category (basal BMI $\geq 25 \text{ Kg/m}^2$). Subjects with an estimated glomerular filtration rate $< 60 \text{ mL/min/1.73 m}^2$ were defined as having a CKD.

Results: After adjusting several covariates, compared with the maintained BMI group, the severe BMI decrease group showed a significantly low risk of CKD development (odds ratio (OR) 0.70, 95% confidence intervals (CI) 0.54-0.91, $p < 0.01$) and the severe BMI increase group had a significantly high risk (OR 1.40, CI 1.08-1.81, $p = 0.01$). A further analysis revealed that the OR of CKD development for the severe BMI increase group in the obesity category was higher than that in the non-obesity category (OR 1.75 vs. 1.29).

Conclusion: In subjects who underwent annual health check-ups, BMI reduction had a significant effect on the prevention of CKD development, whereas an increase in the BMI was a risk factor for CKD development. Moreover, by severe increase in the BMI, obesity subjects showed higher risk of CKD development than non-obesity subjects.

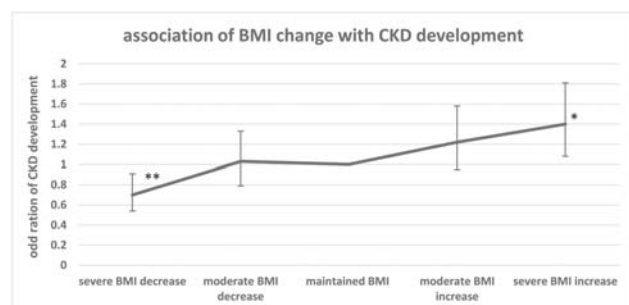


Figure: